Nicola De Lillo

Email: nicoladelillo2001@gmail.com **Phone:** (+39) 3735533389

Date of birth: 15/04/2001 **Nationality:** Italian

ABOUT ME

Computer Engineer with professional experience in software engineering and a strong interest in game development. I am particularly drawn to the intersection between technology and game design, where engineering decisions directly shape gameplay and player experience. During my Master's thesis, I designed a data-driven game architecture and experimented with LLM-driven NPC behaviour in Unity. After gaining valuable professional experience in a large-scale corporate environment, working on production systems and complex software infrastructures, I am now looking to bring my software engineering background into the game industry and further develop my skills in engine and gameplay programming.

WORK EXPERIENCE

University of Sannio

City: Benevento | **Country:** Italy

[01/05/2026 – Current] **Junior Researcher**

Responsible for the design of the first operational prototype of the digital catalogue for research infrastructures, integrating data on infrastructures and services. Contributed to defining an intuitive and functional structure to support the future development of the final digital platform.

Responsible for the implementation of the digital census service by providing technical and operational support to the Directorate General of Regione Campania.

Atlas Reply Roma

City: Rome | **Country:** Italy

[03/10/2025 – 30/04/2026] **DevOps Engineer**

I worked on the migration of CI/CD processes from an on-premises Jenkins environment to Azure DevOps. My responsibilities included designing YAML-based pipelines, implementing environment approval gates, managing secrets, and creating reusable templates. I was also responsible for designing, maintaining, and refactoring the CI/CD pipelines of several microservices and applications, as well as supporting and maintaining Kubernetes clusters during deployment and operational activities.

From a development perspective, I contributed to the creation of an agentic AI platform for incident response. In particular, I implemented MCP servers to securely expose telemetry data, operational tools, and runbooks to LLM-based agents, ensuring controlled and auditable execution. I also worked with Prometheus, Grafana, and several database management systems, developing custom extractors to retrieve and migrate data from legacy on-premises platforms.

In addition, I was responsible for migrating multiple microservices from an on-premises Kafka infrastructure to Confluent Cloud. This involved creating and maintaining deployment pipelines, configuring application values, and supporting the migration from both an infrastructure and software development perspective.

Eutech Engineers

City: Lisbon | **Country:** Portugal

[07/2024 – 09/2024] **Erasmus+ Traineeship**

Awarded an Erasmus+ scholarship for a traineeship abroad; responsible for designing and managing the business logic layer of an e-commerce application, ensuring scalability and maintainability.

Bars S.R.L

City: Benevento | **Country:** Italy

[04/2023 – 06/2023] **Traineeship**

Contributed to the development workflow of augmented reality apps using TeamViewer Frontline, focusing on optimizing processes and supporting application deployment.

PROJECTS

[03/02/2025 – 02/07/2025] **Reve — Game Architecture & AI Systems | Unity, C#**

As part of my master's thesis I've designed and implemented a data-driven game architecture enabling designers to modify rules, item stats and NPC behaviors via editable XML files, with modular systems for combat, items, and AI to support rapid iteration.

As part of the thesis, researched and prototyped the integration of Large Language Models into Unity to orchestrate NPC decisions and dialogue in real time, including a memory system for consistent, context-aware interactions.

The project was tested in a full game prototype made in Unity.

Source code is available on github, while additional technical details, architecture diagrams and gameplay demonstrations are available in the thesis and project presentation.

Link: https://github.com/nik200122/Reve-Project

EDUCATION AND TRAINING

[28/09/2023 – 14/07/2025] **Master's Degree in Computer Engineering**

University of Sannio https://www.unisannio.it/it

City: Benevento | **Country:** Italy | **Field(s) of study:** Information and Communication Technologies | **Final grade:** 110/110 Cum Laude With Honors | **Level in EQF:** EQF level 7 | **Thesis:** An LLM-based architecture for automating NPC actions in videogames

Completed a Master’s in Computer Engineering with a strong focus on software engineering, data science, distributed systems, and game development. Developed advanced skills in designing scalable architectures, optimizing machine learning models, and integrating emerging technologies such as Large Language Models into interactive applications.

Game Development & Thesis Project (Data-Driven Architecture + LLM for NPCs)

Designed and implemented a data-driven game architecture enabling designers to modify rules, item stats and NPC behaviors via editable XML files, with modular systems for combat, items, and AI to support rapid iteration.

As part of the thesis, researched and prototyped the integration of Large Language Models into Unity to orchestrate NPC decisions and dialogue in real time, including a memory system for consistent, context-aware interactions.

Software Engineering Project (SZZ Algorithm Tool)

Developed a software tool implementing the SZZ algorithm to automatically identify bug-introducing commits from version control history, focusing on modularity and maintainability.

Distributed Systems Architecture Project ("Ermes")

Developed the multicast service of "Ermes," an emergency management platform for smart cities, enabling real-time streaming of ambulance alerts to vehicles on a live map so drivers can clear the way promptly.

Data Science Project 1 (Neural Network Hyperparameter Optimization)

Optimized hyperparameters of a Feedforward Neural Network and an Autoencoder to detect DoS attacks with high recall and low false positive rates, and analyzed the performance impact of varying key parameters.

Data Science Project 2 (Big Data Processing and Real-Time Streams)

Implemented MapReduce programs and Spark queries to compute statistics on large visit datasets, and built an Apache Storm topology to sample and trigger alerts on high-temperature events in real-time data streams.

[09/2020 – 09/2023] **Bachelor's Degree in Computer Engineering**

University of Sannio https://www.unisannio.it/it

City: Benevento | **Country:** Italy | **Field(s) of study:** Information and Communication Technologies: • *Inter-disciplinary programmes and qualifications involving Information and Communication Technologies (ICTs)* | **Final grade:** 110/110 Cum Laude With Honors | **Level in EQF:** EQF level 6 | **Thesis:** Streaming and web visualization of traffic speed forecast

The undergraduate program provided me with a comprehensive background in core topics such as data structures, databases, operating systems, and networks, which I later applied to more advanced projects.

Bachelor's Thesis – Real-Time Traffic Forecast Middleware

Designed and implemented a middleware component to stream real-time traffic predictions from Apache Kafka to a web browser, displaying the city of Lyon's map and enabling dynamic, geo-spatial subscription to relevant data streams based on the user's selected viewport.

[09/2015 – 07/2020] **Secondary School Diploma in Scientific Studies**

I.I.S Galilei-Vetrone https://www.iisgalileivetrone.edu.it

City: Benevento | **Country:** Italy | **Field(s) of study:** Scientific Studies | **Final grade:** 100/100 | **Level in EQF:** EQF level 4

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

DevOps

Programming Languages

C | Python | Java | C# | R | C++

Databases and Query Languages

SQL | MongoDB | PostgreSQL | Graph Databases | Oracle Relational Database | MySQL

Distributed Systems

Kubernetes | Azure DevOps Pipelines & Repos | Confluent | OpenShift | Apache Kafka | Git | Jenkins | Docker

Data Visualization

Grafana | Prometheus

Data Engineering & Data Science

Apache Spark | Hadoop | Apache Storm

Game Development

Unreal Engine | Unity (digital game creation systems) | Game Design | Level Design | Gameplay Systems | Game Architecture

CERTIFICATIONS

[Cambridge University Press & Assessment, 01/06/2026]

Cambridge C2 Proficiency

Mode of learning: Presential